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MILITARY SPECIFICATION

INDICATOR, CABIN AIR PRESSURE, PRESSURIZED AIRCRAFT, 1-7/8 INCH DIAL, TYPE MA-2

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department.

1. SCOPE

1.1 Scope. This specification covers one type of single pointer cabin air pressure indicator, designated Type MA-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Sheet and Strip, Thin Gauge Polyolefin
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-601	Boxes, Wood, Cleated Plywood

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier, Material, Greaseproofed, Flexible
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

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STANDARDS

FEDERAL

FED-STD-595	Color

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection Of
MIL-STD-794	Parts and Equipment, Procedures for Packaging And Packing Of
MIL-STD-810	Environmental, Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MS28105	Cover Glass, Aircraft Instrument Dial
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form
MS33585	Pointer, Dial Standard Design of Aircraft Instrument
MS33638	Case, Instrument, Flange-Mounted, Aircraft

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 633 Zinc on Iron and Steel, Electrodeposited Coatings Of

ASTM D 3951 Standard Practice For Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified, a sample shall be subjected to first article inspection (see 4.6 and 6.3).

3.2 Material.

3.2.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.2.2 Materials shall conform to applicable specifications as specified herein. When materials are used which are not specifically designed, they shall be entirely suitable for the purpose. The use of light weight materials and weight saving designs is a major consideration, and their use shall be investigated and exploited to the greatest possible extent.

3.2.3 Metals. Metals shall be of the corrosion resisting type or suitably treated to resist corrosion in fuels, salt spray, or atmospheric conditions to which the system components will be subjected when in storage or during normal service use.

* 3.2.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined by MIL-STD-889.

3.2.5 Nonmagnetic material. Nonmagnetic material shall be used for all parts, except where magnetic materials are essential.

3.2.6 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the procuring activity.

3.2.7 Protective treatment. When materials are used in the construction of the system that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

* 3.2.8 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.3 Design and construction.

3.3.1 The mechanism shall be a pressure actuated type. The design shall be reasonably simple to permit overhaul or repair without requiring numerous special tools and fixtures. Indicators shall be so constructed as to withstand the normal strains of jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service without failure.

3.3.2 Case. The case shall consist of a body and bezel ring and shall conform to MS33638. The over-all length of case and bezel ring shall be 1-3/4 $\pm$1/8 inch.

3.3.2.1 Body. The body shall be made of nonferrous low density metal, shall be uniform in texture, and shall have a smooth surface, with durable black finish.

3.3.2.1.1 Pressure insert. One pressure connection insert shall be located at the rear of the case and centered within 1/8 inch. The insert shall be threaded with 1/8 internal ANPT.

3.3.2.1.2 Reinforcement. The inside of the case around the connection shall be of sufficient strength to prevent damage to the case when the connection is tightened during installation of the indicator.

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3.3.2.2 Bezel Ring. The bezel ring shall be made of nonferrous low-density metal and shall have a durable black finish. The bezel ring shall be held in place by means of screws. Provision shall be made for the replacement of the cover glass by removal of the bezel ring.

3.3.2.3 Mounting lugs. Mounting lugs of the complete assembled case shall each withstand for 1 minute without fracture a load of 175 pounds applied along the mounting hole axis and toward the front of the instrument.

3.3.2.4 Cover glass. The cover glass shall conform to MS28105-7.

3.3.2.5 Shipping plug. A suitable shipping plug shall be screwed in the pressure insert for shipping purposes. The plug shall incorporate a vent consisting of a hole 1/16 ±1/64 inch in diameter.

3.3.3 Mechanism adjustment. The mechanism shall be provided with means of adjusting or correcting the indications to conform to the performance required herein, in case this becomes necessary during the life of the indicator. The calibration adjustment shall not affect the temperature performance of the instrument. This means of adjustment shall be reasonably simple to manipulate and of such nature that adjustment can be accomplished by tools ordinarily possessed by an instrument repairman.

3.3.4 Distance from dial to cover glass. The distance from dial to cover glass shall be as small as practicable and shall not exceed 0.125 inch.

3.3.5 Pointer. The pointer shall be in accordance with MS33585-1.

* 3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.5 Performance. The indicator shall be capable of satisfactory operation when subjected to the following conditions:

a. Temperature. Temperatures ranging from -54° Centigrade (C) to +71°C.

b. Humidity. Relative humidity up to 100 percent.

c. Altitude. Altitudes ranging from 30 inches Hg down on 0.809 inch Hg.

d. Vibration. Vibration incident to service use.

e. Case leakage. A vacuum sufficient to cause a change of reading of the pointer of approximately 18,000 feet.

f. Magnetic effect. Exposure to the magnetic effect of a short bar magnetic compass.

g. Hysteresis. Hysteresis effects as will be encountered in service.

3.6 Markings. All markings shall be durable to withstand usage encountered in service. The form of the letters and numerals on the dial shall conform to Standard MS33558.

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3.6.1 Fluorescent-luminescent material. The following markings shall be finished in lusterless white, color No. 37875 of FED-STD-595.

Markings	Height or Length of Graduation Inches ± 0.010	Width of Line of Graduation Inches ± 0.005
Numerals 0, 10, 20, 30, 40, 50, 60, 70, and 80	0.125	0.020
Graduations Corresponding to "0" and each subsequent 5,000 Foot Graduation.	As shown on Figure 1	0.020
All Other Graduations	As shown on Figure 1	0.012
Lettering "Cabin Pressure Altitude"	0.093	0.016
Lettering "1000 Ft"	0.062	--
Shaded Portion of Pointer	--	--

3.6.2 Durable dull black. The unshaded portion of the pointer and the background of the dial shall be finished in durable dull black.

3.7 Threads.

3.7.1 Screw. Screw threads shall conform to MIL-S-7742.

3.7.2 Pipe. Pipe threads shall conform to MIL-P-7105.

3.8 Weight. The weight of the indicator shall not exceed 0.5 pound.

3.9 Finishes and protective coatings.

3.9.1 Finishes and protective coatings which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions, shall not be used.

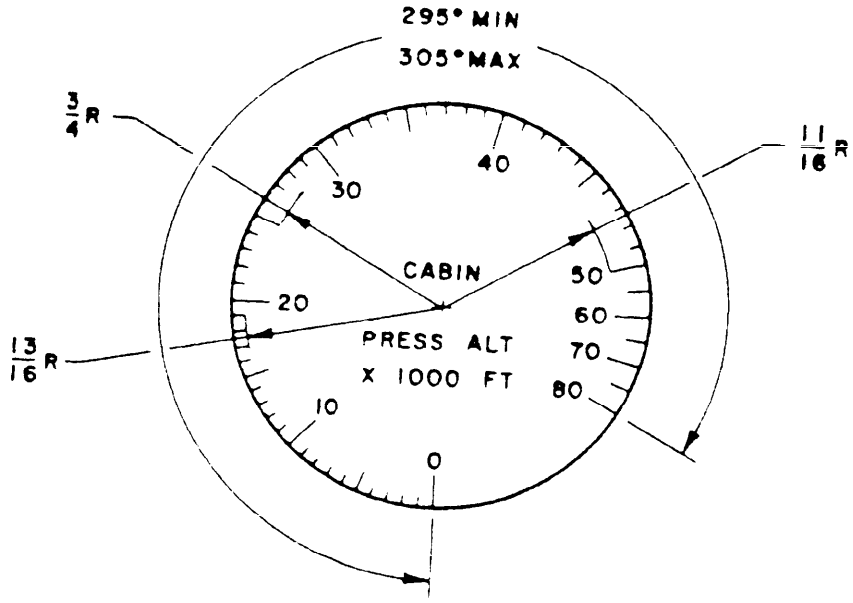
3.9.2 Aluminum alloy parts. Aluminum alloy parts, where practicable, shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, all small holes, pipe threads, and case inserts, need not be anodized.

3.9.3 Steel parts. Steel parts shall be zinc plated where applicable. Zinc plating shall be in accordance with ASTM B 633, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.11 Installation. The contractor shall furnish sufficient mounting screws for installing the indicator. The screws shall be No. 6-32 NC-2A round head brass matching screws, have a dull black finish, and of sufficient length for mounting on a panel 3/8 inch thick.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES:
FRACTIONS ± 1/64
ANGLES ± 0° 15'

FIGURE 1

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3.11.1 Envelope. An envelope containing the mounting screws shall be packaged with each indicator and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS MOUNTING SCREWS"

3.12 Workmanship.

3.12.1 General. The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.12.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with best shop practices. Where dimensions and tolerances affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.12.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "right" means that the screw or bolt cannot be appreciably tightened further without damage or injuty to the screw or bolt threads.

3.12.4 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.12.5 Gears. Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots, or other irregularities. Where required, for accuracy adjustment, gear assemblies shall be free from back lash.

3.12.6 Cleaning. Units shall be thoroughly cleaned of loose, spattered or excess solder, metal chips or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Acceptance tests. Acceptance tests are those tests accomplished on systems submitted for acceptance under contract.

* b. First article inspection (see 4.6).

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4.3 Acceptance tests. The acceptance tests shall consist of the individual tests and those specified under sampling plan.

4.3.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.5 of this specification.

a. Examination of product

b. Case leakage

c. Scale error - room temperature

d. Hysteresis

4.3.2 Sampling plans and tests.

4.3.2.1 Sampling plan. Five indicators shall be selected at random from each lot of 500 produced and subjected to the tests listed below and described under 4.5 of this specification.

4.3.2.2 Lot. A lot shall consist of indicators made under essentially the same conditions and submitted for inspection at about the same time.

a. Scale error (-55°C)

b. Position error

c. High temperature characteristics

d. Vibration error

e. Friction

f. Overpressure

4.3.2.3 Rejection and retest. When one or more items fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be replaced.

4.3.2.4 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all requirements of the specification.

4.4 Test conditions.

4.4.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

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4.4.2 Tapping. Unless otherwise specified, the indicator shall be tested in normal operating position and shall be lightly tapped before a test reading is taken.

4.4.3 Vibration. A vibration shall be used which will vibrate at any desired frequency between 480 Cyc/Min (1 Hz) and 3000 Cyc/Min (50 Hz) and shall subject the indicator to vibration such that a point on the instrument case will describe, in a plane inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

4.5 Test methods.

4.5.1 Examination of product. The indicator shall be inspected to determine compliance with this specification with respect to materials, workmanship, and marking.

4.5.2 Case leakage. A vacuum sufficient to produce a change of reading of the pointer of approximately 18,000 feet shall be slowly applied to the static pressure connection of the indicator at which point the connection tubing shall be pinched off or otherwise completely sealed. The change in reading in 10 seconds shall not exceed 100 feet.

4.5.3 Scale error-room temperature. The indicator shall be subjected successively to the pressures specified in Table I. The reduction in pressure, as indicated by a standard mercurial barometer during the interval between test points, shall be made at a rate corresponding to an increase in altitude of approximately 3,000 feet per minute. The indicator shall remain at the pressure corresponding to each point for at least 1 minute, but not more than 10 minutes before a test reading is taken. The scale errors shall be tabulated. Movement of the point shall be free from back lash and any irregular action when the pressure is varied uniformly. The scale error at any point shall not exceed the tolerance listed in Table I.

4.5.4 Hysteresis. Not more than 15 minutes after the indicator has been first subjected to the pressure corresponding to the upper limit of the scale in the scale error test, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 feet per minute until the pressure corresponding to the first test point given in Table II is reached. The indicator shall remain at this pressure for at least 5 minutes but not more than 15 minutes, before the test reading is taken. After the reading has been taken the pressure shall be further increased at the above rate until the pressure corresponding to the second test point given in Table II is reached. The indicator shall remain at this pressure for at least 1 minute but not more than 10 minutes before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until atmospheric pressure is reached. The reading of the indicator at either of the two points shall not differ from the reading of the indicator for the corresponding altitude in the scale error test by more than the tolerance given in Table II.

4.5.4.1 After effect. Not less than 1 minute and not more than 5 minutes after the completion of the hysteresis test, the pointer shall have returned to its original reading, corrected for any change in atmospheric pressure, within the tolerance given in Table II at the third point.

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TABLE I Test Tolerances

Standard Altitudes	Equivalent Atmospheric Pressure	Scale Error Tolerance (Feet)	
		Room Temperature (Variation from Standard Altitude)	-55°C (Variation from Room Temperature Reading)
Feet x 1000	Inches of Mercury	Feet x 100	Feet x 100
0	29.92	5	5
5	24.89	5	5
10	20.58	5	5
15	16.88	5	5
20	13.75	5	5
25	11.10	10	10
30	8.88	10	10
35	7.04	10	10
40	5.54	10	10
45	4.36	10	10
50	3.44	15	15
60	2.12	20	20
70	1.31	25	25
80	0.81	40	40

TABLE II Hysteresis

Test Points	Standard Altitude (feet)	Equivalent Atmospheric Pressure (In. Hg)	Permissible Variation (feet)
First Point	25,000	11.10	500
Second Point	20,000	13.75	500
Third Point	----	Ground Level	500

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ecify (see tables I and II). coded wire is connected corre

4.5.5 Scale error (-55°C). For a period of not less than 12 hours prior to this test, the indicator shall not have been subjected to any test involving operation at other than atmospheric pressures. The test specified in 4.5.3 shall be repeated with the difference that the temperature of the indicator during this test shall in this case be approximately -55°C. This temperature having been maintained for a period of 3 hours. The test points shall be those given in Table I. The change in scale errors of this test from corresponding scale errors in the room temperature test shall not exceed the tolerance specified in Table I.

4.5.6 Position error. At normal atmospheric pressure, reading shall be taken while the indicator is being tapped lightly in each of several different positions. The change in pointer indication with change in indicator position shall not exceed 200 feet.

4.5.7 High temperature characteristics. The indicator shall be subjected to a temperature of +71°C for a period of 3 hours. It shall then be allowed to cool to a normal temperature, after which it shall be subjected to the tests specified in 4.5.3 and 4.5.2. The indicator shall meet the requirements specified for these tests.

4.5.9 Vibration error. The indicator shall be mounted on a vibration stand with a line joining the centers of the two upper mounting holes in a horizontal plane and with the plane of the mounting lugs of the indicator vertical. The indicator shall be subjected to vibration with circle diameter between 0.018 and 0.020 inch at frequencies from 500 to 3,000 cycles per minute in order to determine that the natural frequency of the indicator does not occur in this frequency range. While the indicator is being vibrated at frequencies from 500 to 2,500 cycles per minute, the range of pointer oscillation shall not exceed 250 feet and the pointer variation from its original position shall not exceed 250 feet. While the indicator is being vibrated at frequencies from 2,500 to 3,000 cycles per minute the range of pointer oscillation shall not exceed 400 feet and the pointer variation from its original position shall not exceed 400 feet.

4.5.10 Friction. The indicator shall be tested for friction at each of the test points listed in Table III. The pressure shall be decreased to bring the pointer approximately to the desired reading, then held constant while two readings are taken; the first before the indicator is tapped, the second , after tapping. The difference of the two readings shall be recorded as friction and shall not exceed the tolerances specified in Table III. This test may be combined with the test specified in 4.5.3.

4.5.11 Overpressure. The indicator shall be subjected to an absolute pressure of 50 inches Hg for a period of 1 hour. For a period of not less than 6 hours following this overpressure, the indicator shall not be subjected to any operation other than atmospheric pressure. The scale errors shall not vary from the previously noted scale errors by more than the tolerances listed in Table IV.

4.6 First Article Tests - The first article tests shall consist of the following tests and those specified under test methods.

4.6.1 Magnetic effect. The indicator shall be held in various positions with its nearest part 5 inches from and magnetically east or west of the center of a short bar magnet compass. The compass deflection shall not exceed 1 degree. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted.

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TABLE III Friction

Altitude	Tolerance
Feet x 1000	Feet x 100
5	5
10	5
15	5
20	5
25	5
30	5
35	5
40	5
50	5
60	10
70	15
80	15

TABLE IV Variation from Original Scale Errors After Subjection to 50 Inches Hg Pressure

Stand Alt. Feet x 1000	Variation Feet x 100
0	5
5	5
10	5
20	5
25	5
30	5
35	5
40	5
45	5
50	5
70	10
80	10

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4.6.2 Vibration endurance. The scale error shall be noted at the existing atmospheric pressure. The instrument shall be subjected to vibration with a circle diameter of 0.020 inch for a 3-hour period at a frequency of 2000 cycles per minute. At the end of the vibration period the scale error at atmospheric pressure shall be again noted. The difference between the reading determined after vibration, after correction for any changes in atmospheric pressure, shall not exceed 100 feet.

4.6.2.3 High altitude - low temperature. The indicator shall be placed in a chamber and maintained at a temperature of approximately -65°C and a pressure equivalent to approximately 80,000 feet (0.809 inch Hg) for 4 hours. The indicator shall then be maintained at a temperature of approximately -65°C (atmospheric pressure) for an additional period of 44 hours. There shall be no damage which would affect subsequent operation.

4.6.2.4 Humidity. The indicator shall be subjected to a humidity test in accordance with procedure I of MIL-STD-810, except for a period of 5 cycles. Within 1 hour after completion of the last cycle, a scale error at room temperature test shall be conducted. The scale error shall not exceed the tolerances specified in Table I. Between 24 and 48 hours after completion of the last cycle, the mechanism of the indicator shall be examined for corrosion. There shall be no evidence of corrosion of the internal or external parts of the indicator which will affect the performance of the instrument.

4.6.2.5 Mounting lugs. The indicator case, with mechanism removed, shall be mounted face downward on the movable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug or mounting nut and attached to the pull strap. A load of 175 pounds shall be applied along the axis toward the front of the case for a period of one minute. There shall be no fracture.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the indicators shall be inspected to determine conformance with the requirements of Section 5 of this specification.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A, C or commercial as specified (see 6.2).

* 5.1.1 Level A. Indicators shall be preserved and packaged Method 1C of MIL-P-116. Quantity per unit pack (QUP) shall be one each. Each indicator shall be wrapped in polyolefin film conforming to L-P-378 or equal, adequately cushioned and encased in a heat sealed bag conforming to MIL-B-121, Grade A or equal. Each indicator shall then be placed in a fiberboard box conforming to PPP-B-636, Type CF, Class Domestic. Intermediate packaging when required shall conform to the requirements of MIL-STD-794.

* 5.1.2 Level C. The Level C preservation for indicators shall conform to the MIL-STD-794 requirements for this level.

* 5.1.3 Commercial. The commercial preservation of the indicators shall conform to the requirements of ASTM D 3951.

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* 5.2 Packing. Packing shall be Level A, B, C or Commercial, as specified (see 6.2).

* 5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-601, overseas type. The closure shall be in accordance with the appendix of the container specification.

* 5.2.2 Level B. Indicators preserved and packaged as specified in 5.1.2 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Grade and weight limitations of shipping containers shall be governed by the applicable special requirements table of the specification. Where practical, shipping containers shall be of uniform shape and size, of minimum cube and contain identical quantities. Closure and strapping shall be in accordance with the appendix to PPP-B-636.

* 5.2.3 Level C. Same as level B except Class Domestic shipping containers shall be used.

* 5.2.4 Commercial. The preserved indicator shall be packed in accordance with ASTM-D-3951.

* 5.3 Marking.

* 5.3.1 Levels A, B and C. In addition to any special or other identification marking required by the contract (See 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

* 5.3.2 Commercial. Commercial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The Type MA-2 indicator covered by this specification is intended for use in aircraft equipped with pressurized cabins to indicate the pressure altitude in the cabins.

* 6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following: The level of packaging and packing desired (see 5.1).

* 6.3 First Article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

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6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
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San Antonio Air Logistics Center
Kelly AFB, TX 78241

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-25091B(USAF)	Indicator, Cabin Air Pressure, Pressurized Aircraft, 1-7/8 Inch Dial, Type MA-2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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